

ROBERTSON LAKE, NU, Canada

WMO: 714900

Lat: 65.100N

Lon: 102.430W

Elev: 244

StdP: 98.43

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.1	-40.7	-46.1	0.0	-41.9	-44.5	0.0	-40.4	16.9	-24.3	16.0	-24.0	4.6	260	1.088

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	23.8	14.0	21.2	13.5	18.4	12.5	15.6	21.4	14.3	19.3	12.9	17.4	6.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	9.6	17.1	11.7	8.8	16.1	10.5	8.1	14.8	44.4	21.4	40.7	19.2	37.0	17.5	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	13.8	12.3	-43.3	27.3	1.9	2.6	-44.7	29.1	-45.7	30.6	-46.8	32.0	-48.1	33.9		
(5)			-43.2	18.1	2.0	2.4	-44.6	19.9	-45.8	21.3	-46.9	22.7	-48.4	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.9	-30.8	-30.3	-25.6	-16.5	-6.7	6.2	12.0	9.8	2.8	-7.4	-19.0	-26.8
	DBStd	16.33	6.53	6.66	7.34	6.99	5.59	4.70	3.94	3.65	4.44	5.44	6.23	6.67
	HDD10.0	7782	1266	1128	1104	795	518	133	24	49	218	538	869	1140
	HDD18.3	10678	1524	1361	1362	1045	777	364	199	267	465	797	1119	1398
	CDD10.0	149	0	0	0	0	0	19	86	41	3	0	0	0
	CDD18.3	3	0	0	0	0	0	1	2	1	0	0	0	0
	CDH23.3	81	0	0	0	0	0	14	56	11	0	0	0	0
	CDH26.7	9	0	0	0	0	0	2	5	2	0	0	0	0
Wind	WSAvg	6.1	6.3	6.2	6.2	6.2	6.0	5.2	5.6	6.1	6.7	6.9	6.1	6.2
Precipitation	PrecAvg	230	7	6	11	11	13	20	38	41	36	25	13	9
	PrecMax	281	20	17	27	25	28	40	67	67	62	43	26	18
	PrecMin	154	0	1	2	2	3	7	14	19	18	4	3	1
	PrecStd	30	4	4	6	6	6	10	12	13	12	9	6	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.0	-11.8	-6.0	0.4	6.0	25.1	27.5	24.6	17.9	4.9	-2.6	-7.1
		MCWB	-8.1	-12.0	-6.2	-0.1	4.0	14.4	15.7	14.1	12.9	4.1	-2.7	-7.5
	2%	DB	-13.8	-14.6	-8.8	-2.0	3.6	20.7	25.1	21.1	14.3	2.3	-5.0	-10.4
		MCWB	-14.0	-14.9	-9.0	-2.6	2.4	13.2	14.4	13.0	10.6	1.7	-5.2	-10.6
	5%	DB	-17.8	-16.6	-11.6	-3.9	2.2	17.2	22.9	18.5	11.5	1.0	-7.1	-13.5
		MCWB	-18.1	-16.9	-11.9	-4.2	1.4	11.7	14.1	12.4	8.9	0.7	-7.3	-13.7
	10%	DB	-21.1	-18.9	-14.1	-6.0	0.9	14.4	20.3	16.2	9.0	-0.3	-9.1	-16.3
		MCWB	-21.3	-19.1	-14.3	-6.4	0.3	10.0	13.3	11.2	7.5	-0.6	-9.3	-16.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.4	-11.8	-6.0	0.1	4.5	16.2	17.6	16.1	13.5	4.4	-2.5	-7.7
		MCDB	-9.4	-11.7	-5.9	0.4	5.7	22.2	24.9	21.8	17.2	4.7	-2.5	-7.6
	2%	WB	-14.5	-14.7	-8.8	-2.5	2.5	13.9	16.2	14.4	11.0	2.0	-4.9	-11.6
		MCDB	-14.3	-14.5	-8.6	-2.1	3.3	19.9	22.4	19.0	13.9	2.3	-4.7	-11.5
	5%	WB	-18.3	-16.6	-11.4	-4.2	1.3	12.2	15.1	13.0	9.3	0.7	-7.0	-14.0
		MCDB	-18.1	-16.4	-11.3	-3.8	2.0	17.4	20.7	16.8	11.3	1.0	-6.9	-13.7
	10%	WB	-21.5	-18.7	-13.9	-6.4	0.2	10.5	13.8	11.8	7.7	-0.6	-9.0	-16.9
		MCDB	-21.4	-18.5	-13.8	-6.0	0.7	14.4	18.6	15.3	9.1	-0.3	-8.9	-16.7
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.0	8.0	8.8	6.8	9.5	10.3	8.4	6.0	5.3	6.7	6.8
		MCDBR	10.5	9.5	8.6	9.7	7.0	15.5	15.2	13.2	11.0	5.2	7.8	9.5
		MCWBR	10.5	9.1	8.7	9.4	6.0	9.6	6.8	6.9	7.4	4.9	7.9	9.6
	5% WB	MCDBR	10.4	9.1	8.7	9.9	6.6	14.7	13.2	11.0	10.1	5.3	7.7	9.6
		MCWBR	10.4	9.1	8.6	9.5	5.8	10.1	7.3	6.8	7.2	5.1	7.7	9.6
Clear-Sky Solar Irradiance	taub	0.146	0.227	0.246	0.282	0.302	0.328	0.345	0.316	0.297	0.241	0.157	0.142	
	taud	1.911	2.139	2.256	2.193	2.245	2.453	2.441	2.501	2.520	2.279	1.914	1.926	
	Ebn at Noon	423	687	854	890	903	887	857	855	797	666	366	146	
	Edh at Noon	21	55	79	108	115	96	95	81	64	47	20	6	
All-Sky Solar Radiation	RadAvg	0.15	0.92	2.75	5.22	6.59	5.68	5.21	3.73	2.05	0.65	0.23	0.04	
	RadStd	0.01	0.03	0.12	0.17	0.42	0.42	0.44	0.36	0.25	0.14	0.03	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page